

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
6	(a)			Magnesium - chlorophyll (1) NOT chloroplast Calcium – (structure of) cell <u>walls</u> in plants (1) Phosphate - nucleic acids/ nucleotides/ phospholipids/ ATP/ NADP/ NAD/ FAD(1)	3			3		
	(b)	(i)		Active transport (1) Against concentration gradient (1) Correct use of data (1)	1	2		3		
		(ii)		A. anaerobic conditions / lack of oxygen/ oxygen is required(1) B. {Less/ no} ATP produced (1) C. active transport {cannot occur/ reduced}, fewer ions can be transported (against concentration gradient) (1) D. Stunted growth because of a lack of named ion {e.g. phosphate, required to make DNA and proteins essential for growth} (1)		3	1	4		
				Question 6 total	4	5	1	10	0	0